CT026-3-2-HCI & HUMAN COMPUTER INTERACTION

TOPIC 1: INTRODUCTION TO HUMAN COMPUTER INTERACTION

TOPIC LEARNING OUTCOMES

At the end of this topic, you should be able to:

1. Explain what HCI is
2. Explain what are the factors influencing HCI
3. Explain what usability is
4. Describe the major principles in designing for Usability
5. Explain what interactivity is, and how usability is important in designing interactive product

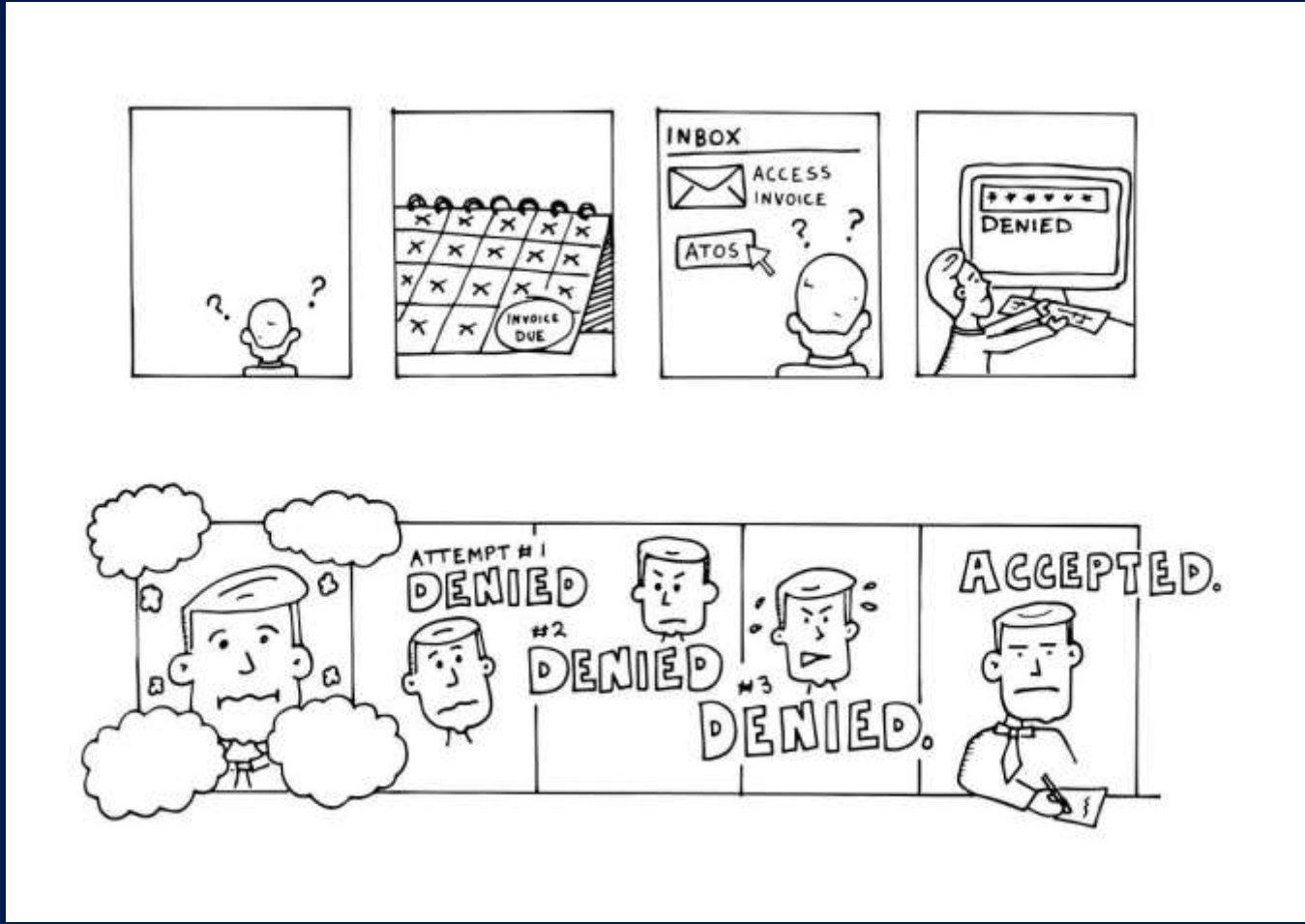
Contents & Structure

1. Common issues in interface design
2. The reason behind HCI
3. HCI definition
4. The many scopes of HCI
5. Usability
6. Designing for usability
7. Understanding interaction

Recap From Last Lesson

- NONE

WHY BOTHER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DENIED.
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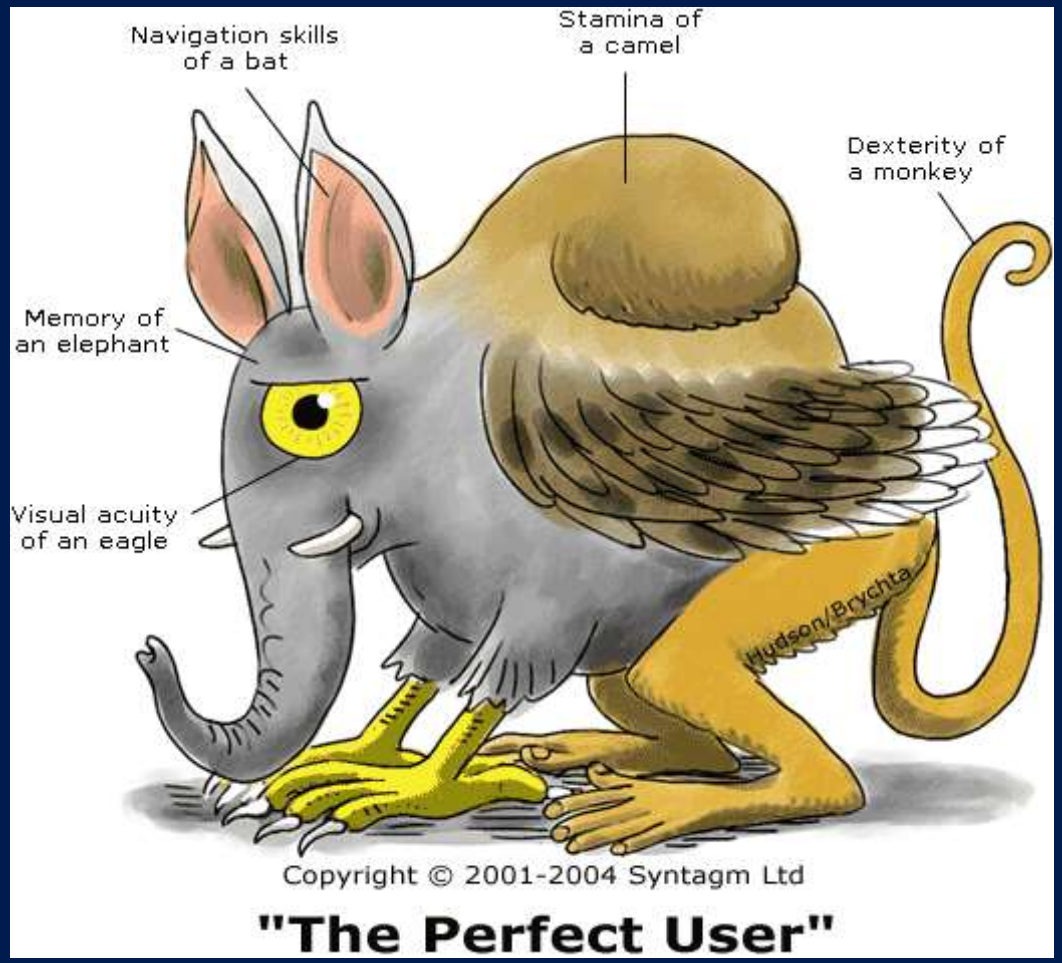
WHY SYSTEM FAIL?

- Common issues / reason on why system fail:
  - Overloaded user interface
    - When the UI of a digital product has too many features, it indicate lack of clarity about the brand's strength.
    - It's overwhelmed the users rather than excitement towards the digital product. User may wonder the company brand is all about.
  - Innovation at the Price of Usability
    - Let's look at eCommerce website.
    - It is crucial for eCommerce websites to follow universal UX rules to ensure a positive user experience. For instance, users expect to find the shopping cart in the upper right-hand corner of the website. Deviating from this rule for the sake of innovation can lead to confusion and frustration for users, ultimately resulting in site abandonment. Therefore, it is important to avoid changing backbone elements that facilitate fast navigation for users and instead focus on innovative design elements that enhance the user experience without compromising usability.
  - Confusing apps

WHY SYSTEM FAIL?

- **Lack of understanding of users' needs**
  - Building great products, including mobile apps, starts with understanding your users.
- **Confusing layouts**
  - Users' persistence with an application can be significantly impacted by its clarity and ease of use. Confusing or complex layouts often result in users giving up their engagement and seeking alternative options due to difficulty in navigating further within the app. Ensuring intuitive designs and clear instructions are vital to retain users and prevent loss of interest.
- **Completely imitating competitors**
- **Designing for yourself instead of the user**
  - A good designer separates his or her personal preferences from the preferences of users. Designing for specific users with specific needs and wants is crucial. Throughout the entire development process, it is crucial to consider users' needs to prevent the app from evolving solely based on the developers' preferences.
- **Confusing UI with UX**
  - UI and UX are distinct but interconnected. One cannot work on a design concept without considering the other.
  - User experience (UX) design, with its emphasis on careful analysis, plays a pivotal role in ensuring the app's functionality is seamless and intuitive for users.
  - Conversely, user interface (UI) design focuses on crafting visually engaging and interactive interfaces. App success relies on effectively utilizing both of these design disciplines in tandem.

THE PERFECT USER – EVERY DESIGNER DREAMS



COMMON ISSUES IN USER INTERFACE DESIGN

- Not optimizing across different devices
- Cluttered navigation
- Failing to address user needs
- Not testing the design enough
- Designing for yourself
- Measuring the presence of UI design principles and many more

WHY STUDY HCI?

Business view :

- To **employ** people more productively and effectively
  - people costs now far outweigh hardware and software costs
- People now **expect** “easy to use” systems
  - generally, they are not tolerant of poorly design the systems
  - if a product is hard to use, they will seek other products

WHY STUDY HCI?

Human Factors view :

- Humans have limitations. Errors are costly in terms of
  - loss of time & money
  - loss of lives in critical systems
  - loss of morale

WHAT IS HCI

~ Short for ***h**uman-**c**omputer **I**nteraction* ~

A discipline concerned with the study, design, construction and implementation of human-centric interactive computer systems.

SCOPE OF HCI

- Human-Computer Interaction (HCI) is a multidisciplinary field that focuses on the design of computer technology and the interaction between humans (users) and computers. HCI addresses a wide array of topics, including:
  - Usability components
  - Design principles
  - Prototyping methods
  - Evaluation
  - Accessibility
  - Decision-making
  - Safety
  - Innovation
  - Context awareness
  - Ethical considerations
- As technology continues to advance, HCI expands to incorporate emerging technologies and user experiences, such as virtual and augmented reality, wearables, and artificial intelligence

THE GOAL OF HCI

The goals of HCI are to develop or improve the safety, utility and effectiveness of systems that include computers, often through improving usability.

WHAT IS USABILITY

- Usability can simply be thought of as the practical implementation of good HCI, but, more formally :
- **Usability means *easy to learn*, *effective to use* and providing an *enjoyable experience***

"It's the latest innovation in office safety. When your computer crashes, an air bag is activated so you won't bang your head in frustration."

HOW TO DESIGN AND BUILD USABLE Uis?

- UI Development process :
  - User Profiling
  - Usability goals
  - Task analysis & contextual inquiry
  - Prototyping
  - Evaluation
  - Hand over for programming

IMPORTANT!!

- Users should be involved throughout the development of the project (How?)
- Specific usability and user experience goals need to be identified, clearly documented and agreed at the beginning of the project

UNDERSTANDING INTERACTION

- User centric design is the formula for usability
- The key to User-centered Design is to understand Interaction
- We need to understand :
  - What Interaction is
  - What are the elements involved

INTERACTION MODEL

- The most influential model of interaction is Donald Norman's (http://www.jnd.org/) :
  -
    - **Execution-Evaluation cycle**
- Norman divides interaction into :
  - **Execution**
    - User activities aimed at making the system do something
  - **Evaluation**
    - Evaluating whether the system did actually do what they wanted

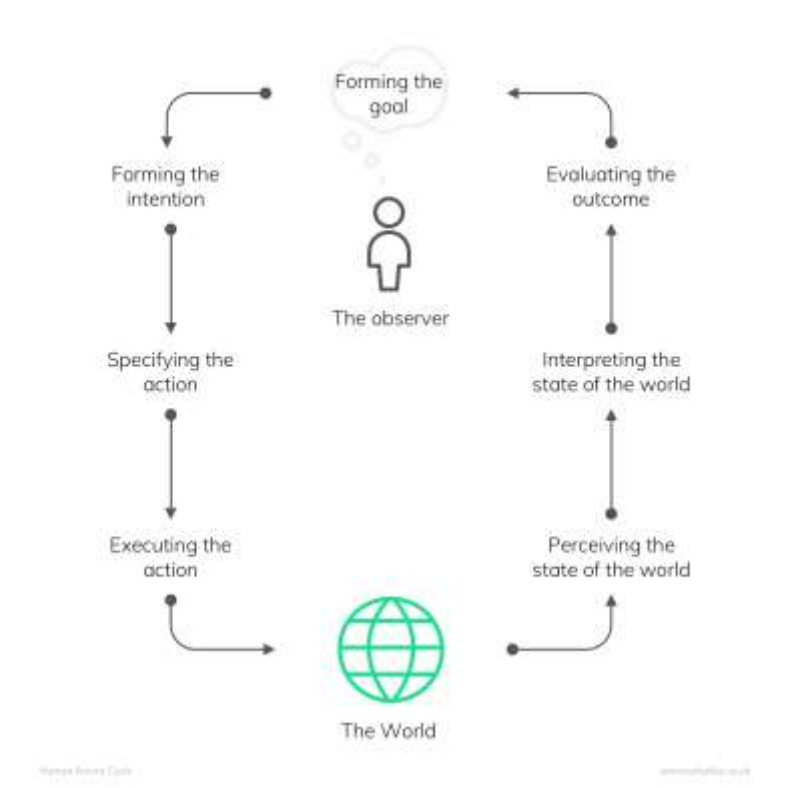


Figure : Interaction model

UNDERSTANDING INTERACTION

- **Execution**
  - If User cannot make system do what they want
    - e.g. cannot understand how to do it, unclear icons, unclear indication etc.
  - Will result in the Gulf of Execution
    - i.e. difference between the user's formulation of the action and the actions allowed by the system

UNDERSTANDING INTERACTION

- **Evaluation**
  - If user cannot see what happened to system
    - e.g. if system has done what they want but no feedback is given to the users etc.
  - Will result in the Gulf of Evaluation
    - i.e. difference between the representation of the system state/result and the expectations of the user

Good Design aims to reduce these gulfs

DIMENSION OF USABILITY

- Does the user feel in control of the application ?
- To what extent can the user achieve their goals using the application ?
- How far does the product appear to assist the user ?
- How easy is the application to learn ?
- How does the user respond emotionally ?

Review Questions

- How can UI designers avoid common mistake in their design?
- How do common mistakes in UI design, such as inconsistent design and lack of contrast, impact the user experience and usability of a digital product?

Summary / Recap of Main Points

- Common issues in interface design
- The reason behind HCI
- HCI definition
- The many scopes of HCI
- Usability
- Designing for usability
- Understanding interaction

What To Expect Next Week

In Class

- HUMAN FACTORS

Preparation for Class